MLOPS ASSIGNMENT

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1. Which run had the lowest latency, and why might that be?

The run_batch8 experiment is expected to have the lowest latency compared to run_batch1. This is because batch processing allows the model to process multiple sentences together instead of processing them individually. When the batch size is increased, the number of processing operations can be reduced, which may improve inference speed and overall efficiency. However, the actual latency depends on the system hardware and available resources.

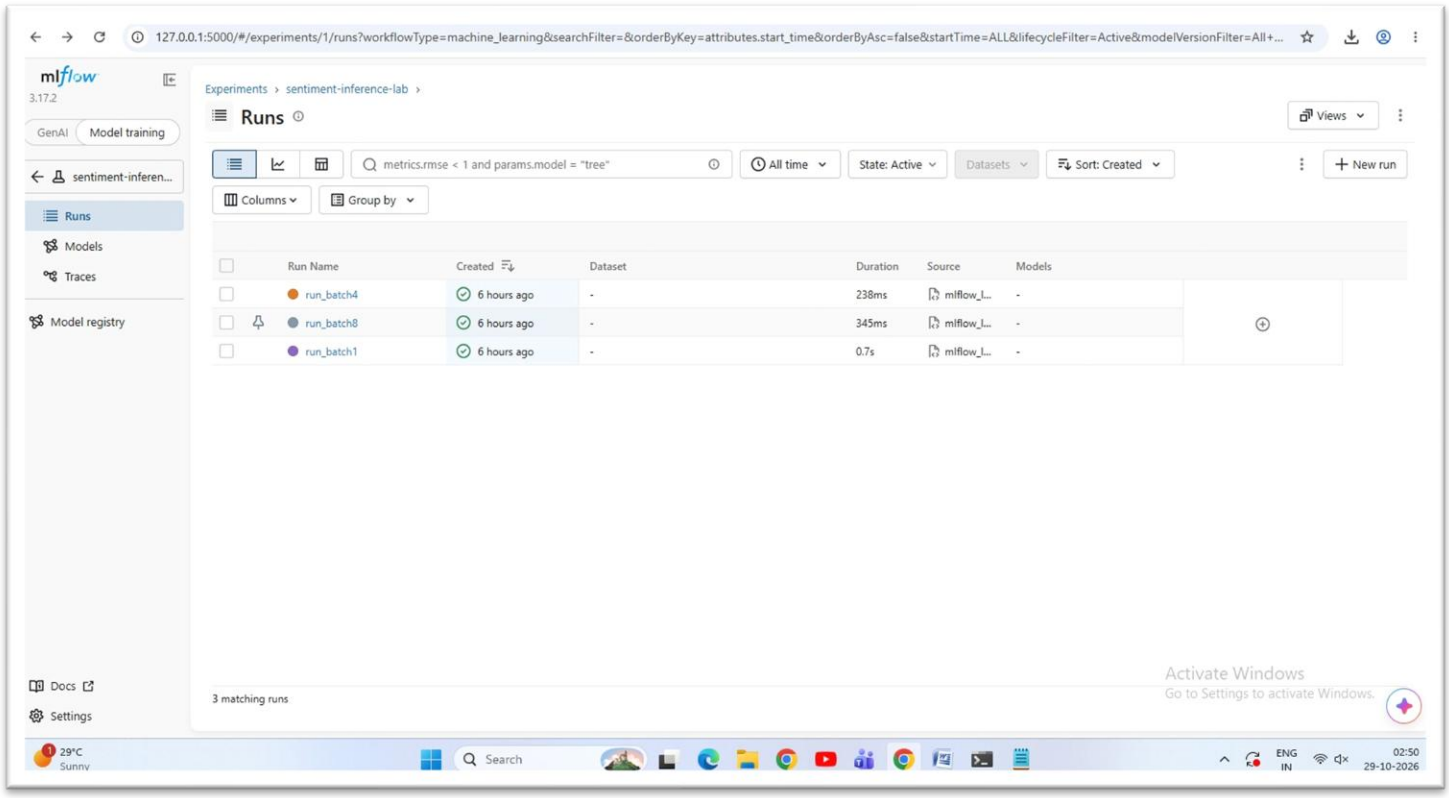
2. Did accuracy change between runs? Why or why not?

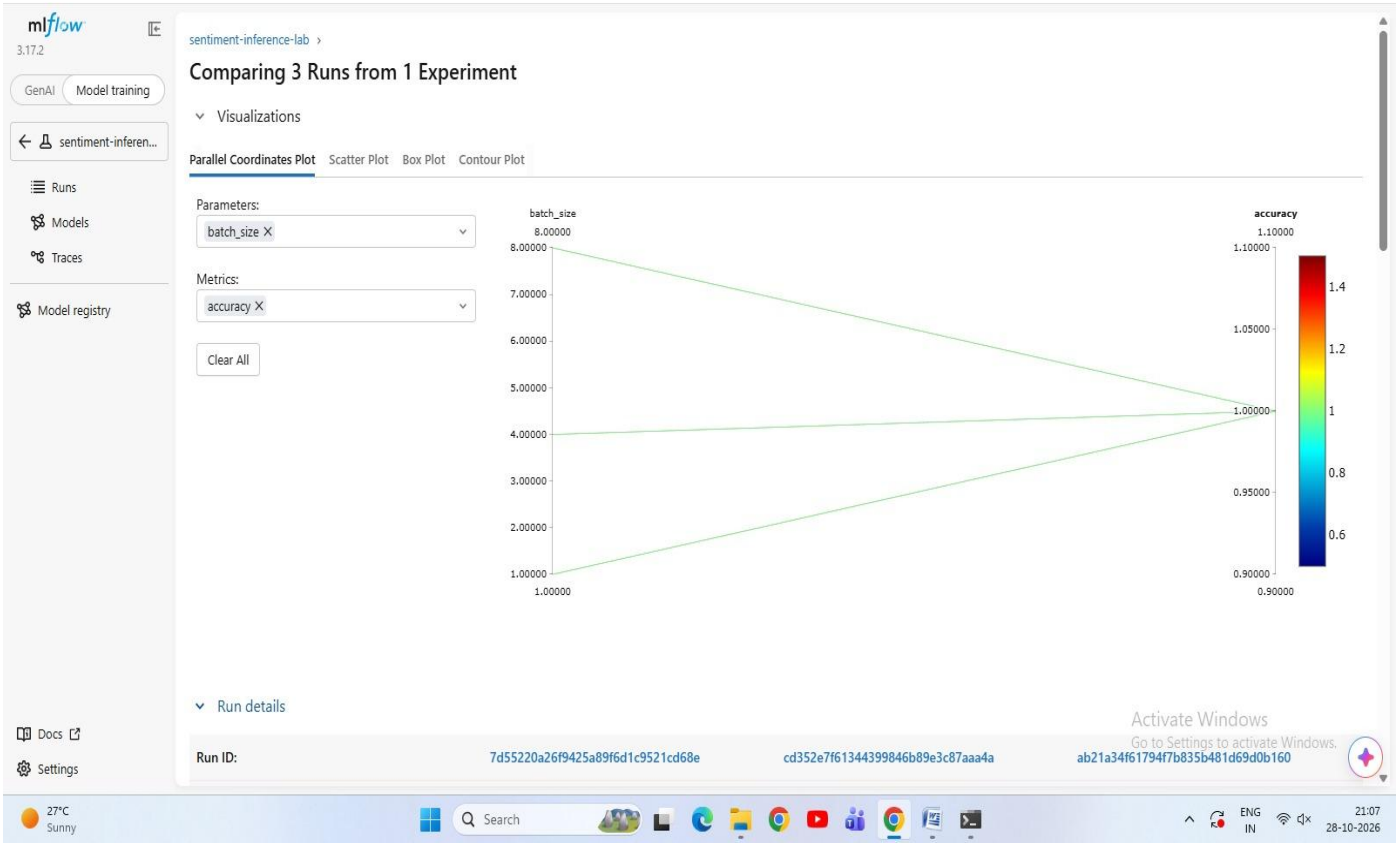
No, the accuracy is expected to remain the same across the runs because the same pre-trained Hugging Face sentiment analysis model (distilbert-base-uncased-finetuned-sst-2-english) is used with the same dataset. Only the batch size is changed between experiments. Batch size affects how the input sentences are processed, but it does not change the model's learned parameters or prediction logic. Therefore, the predicted sentiment labels and accuracy are generally expected to remain unchanged.

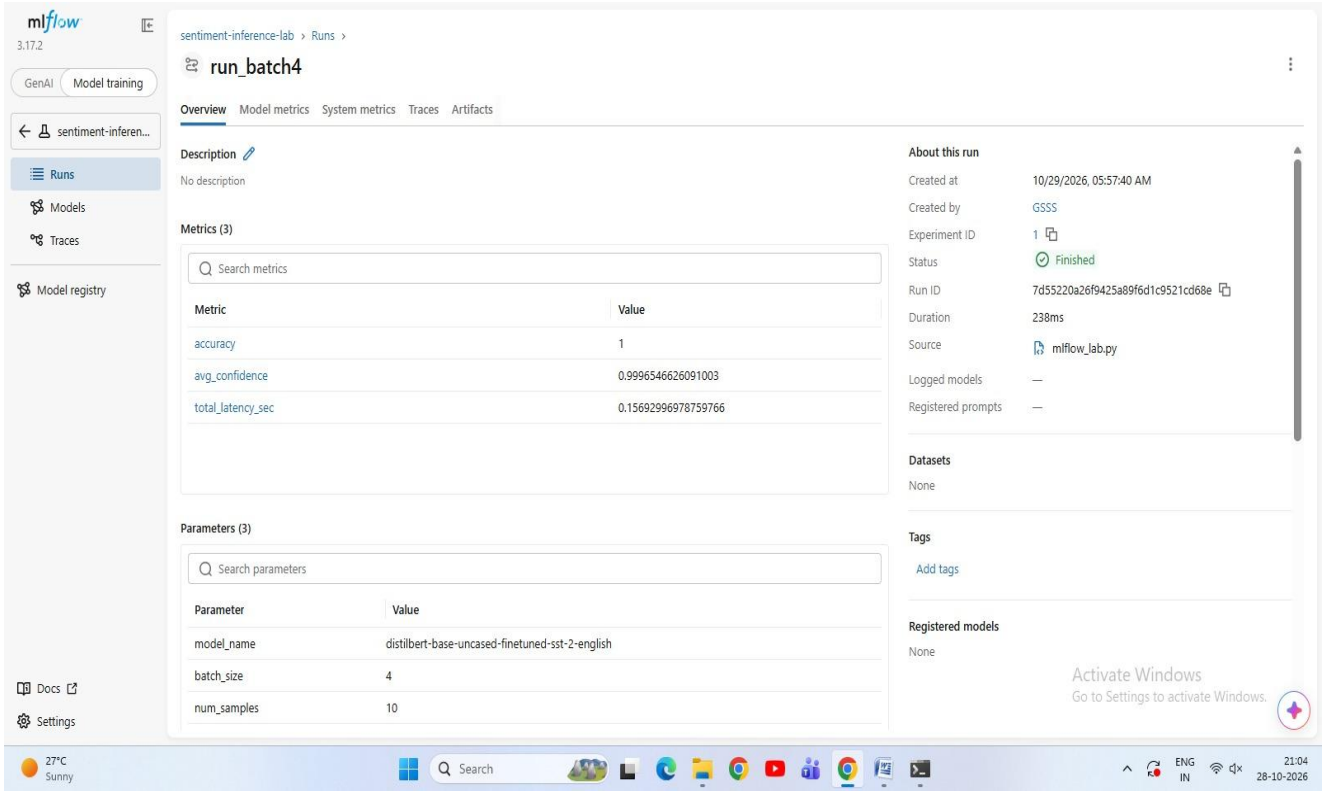
3. Name two benefits of using MLflow here instead of printing results in the terminal.

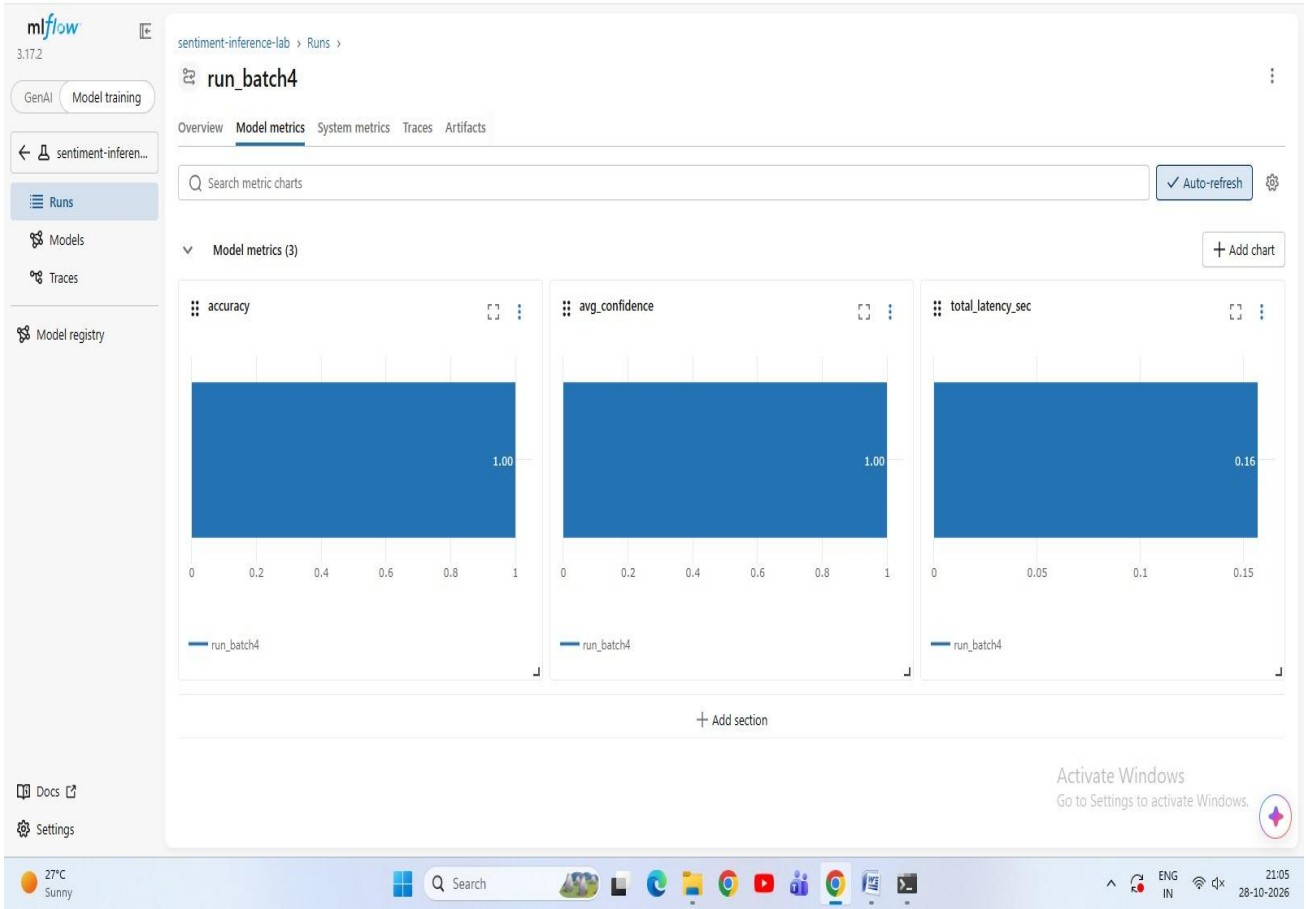
- Experiment Tracking: MLflow automatically records important experiment information such as model name, batch size, number of samples, accuracy, average confidence, and total latency. This makes it easier to maintain and track multiple experiments without manually recording the results.
- Experiment Comparison and Artifact Management: MLflow provides a user-friendly web interface where we can compare multiple experiment runs side by side. It also stores artifacts such as predictions.csv, allowing us to examine the actual and predicted sentiment labels along with their confidence scores. This helps us analyze model performance and identify the better configuration.

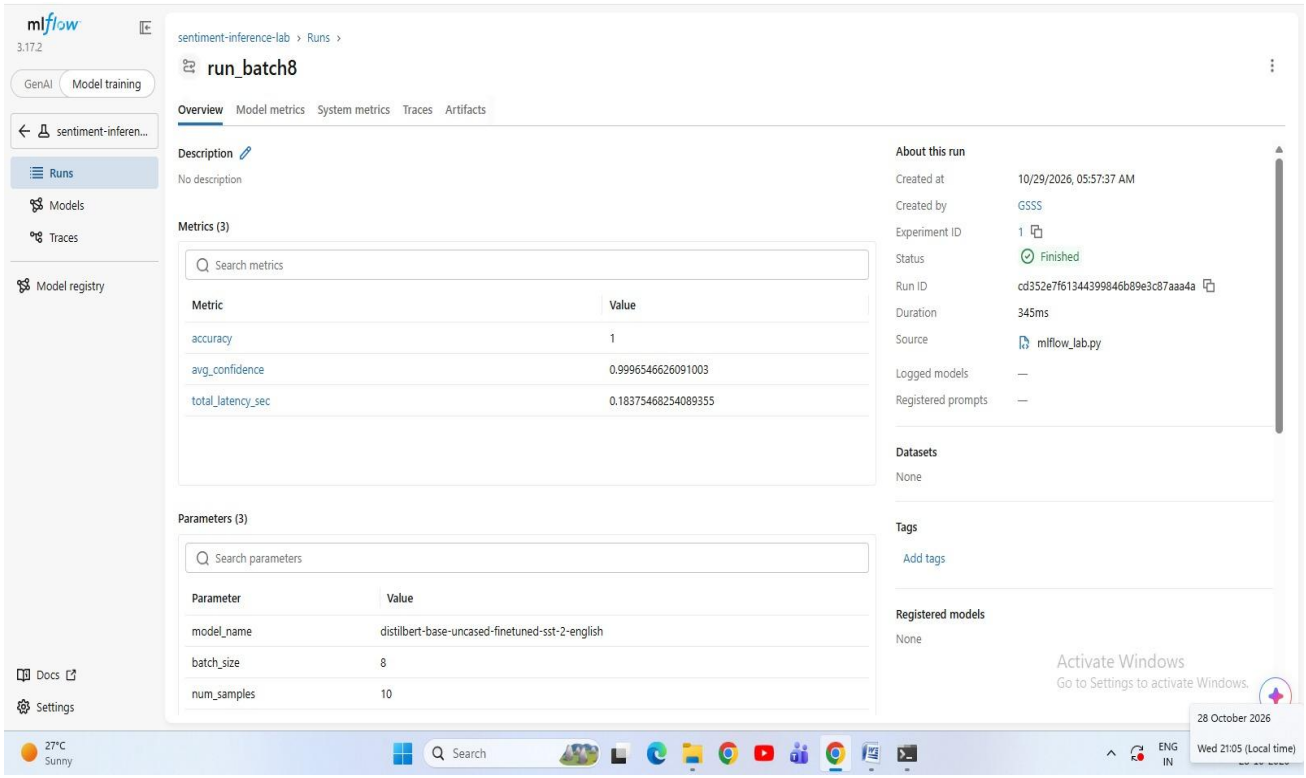
Screenshot of the ML flow

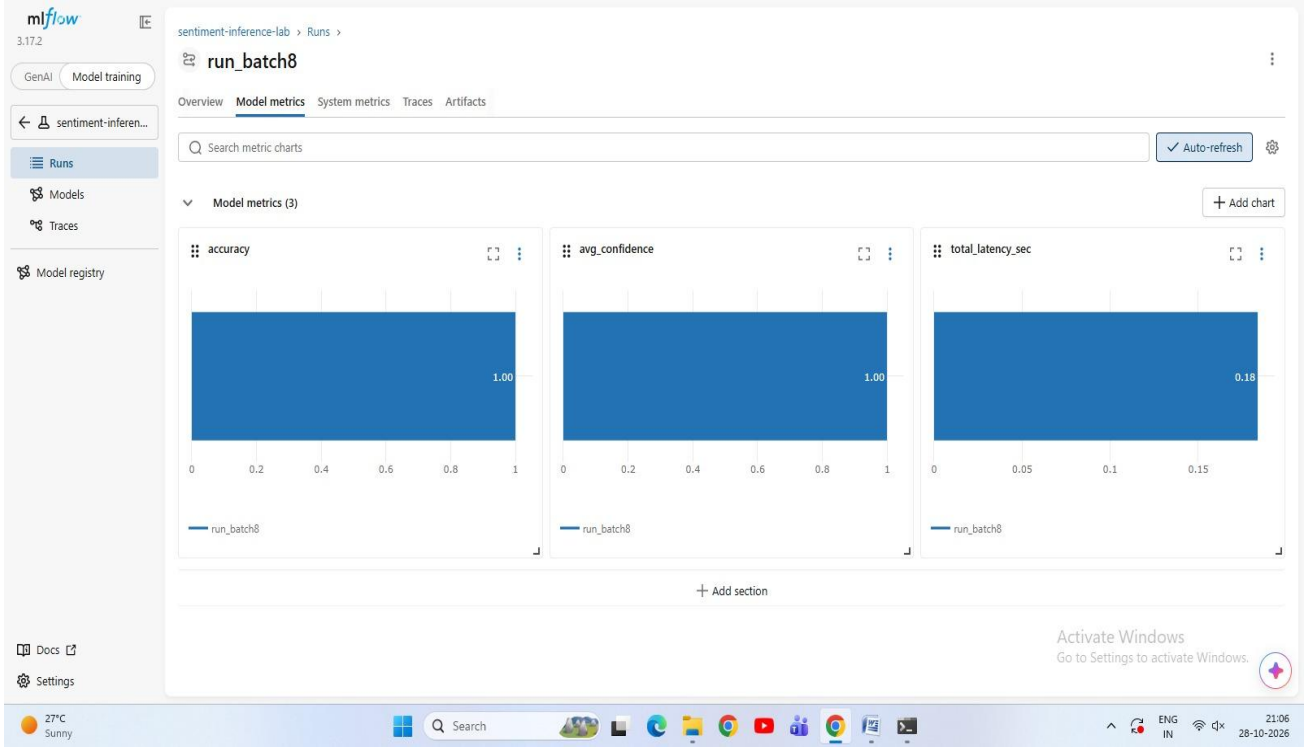


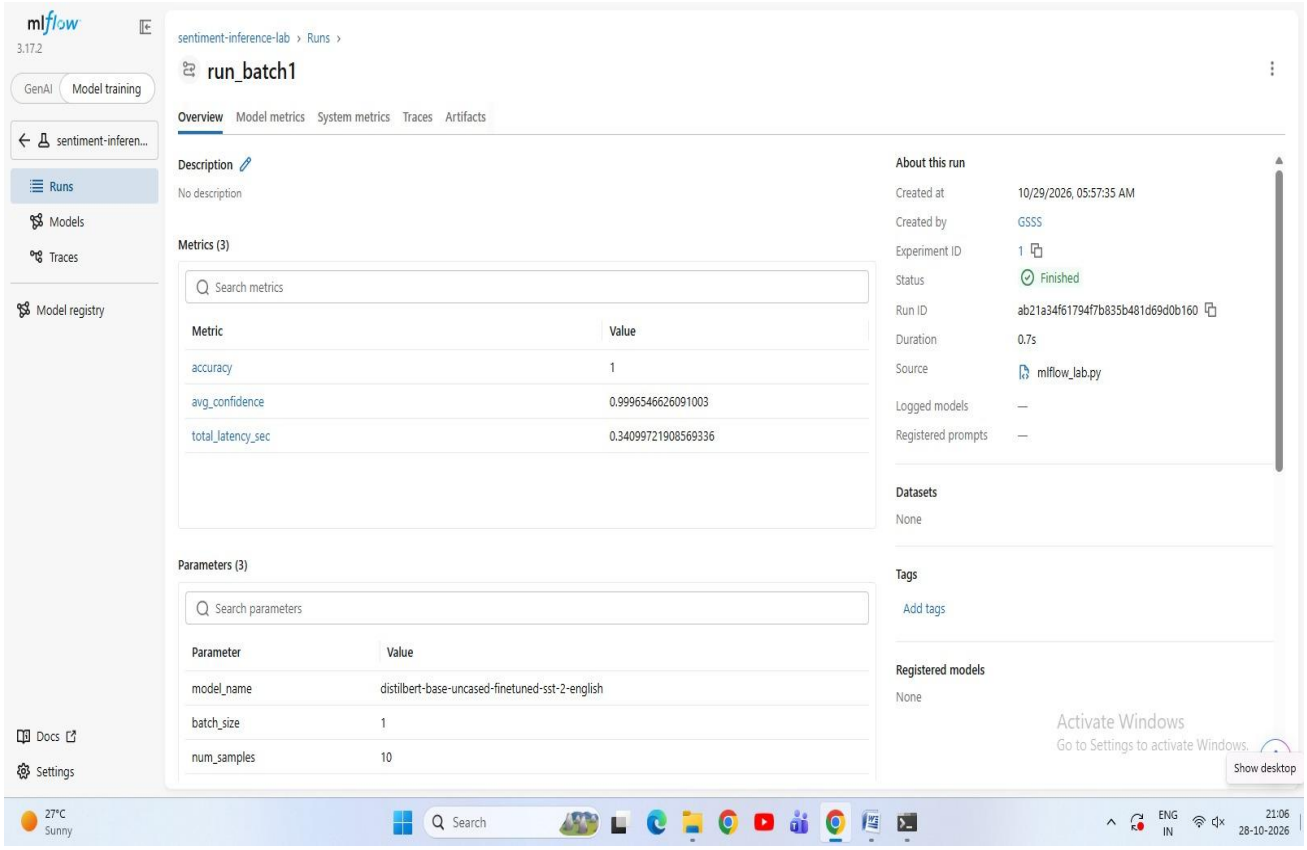


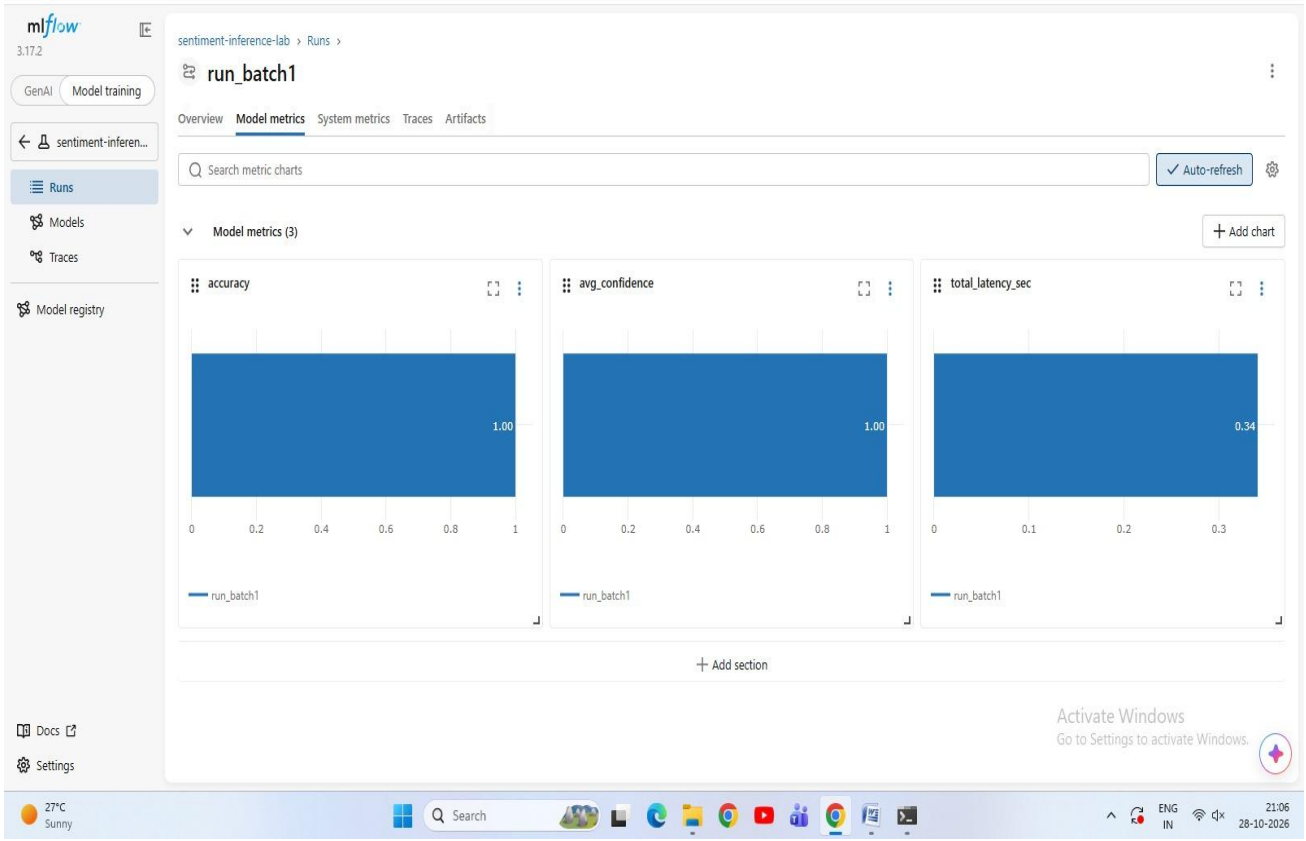

mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch4
Overview
Model metrics
System metrics
Traces
Artifacts
Description
No description
Metrics (3)
Search metrics
Metric
Value
accuracy
1
avg_confidence
0.9996546626091003
total_latency_sec
0.15692996978759766
Parameters (3)
Search parameters
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
4
num_samples
10
About this run
Created at
10/29/2026, 05:57:40 AM
Created by
GSSS
Experiment ID
1
Status
Finished
Run ID
7d55220a26f9425a89f6d1c9521cd68e
Duration
238ms
Source
mlflow_lab.py
Logged models
—
Registered prompts
—
Datasets
None
Tags
Add tags
Registered models
None
Activate Windows
Go to Settings to activate Windows.
27°C
Sunny
Search
ENG
IN
21:04
28-10-2026


mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch4
Overview
Model metrics
System metrics
Traces
Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
1.00
1.00
0.16
0
0.2
0.4
0.6
0.8
1
0
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0.15
run_batch4
Add section
Activate Windows
Go to Settings to activate Windows.
27°C
Sunny
Search
ENG
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21:05
28-10-2026

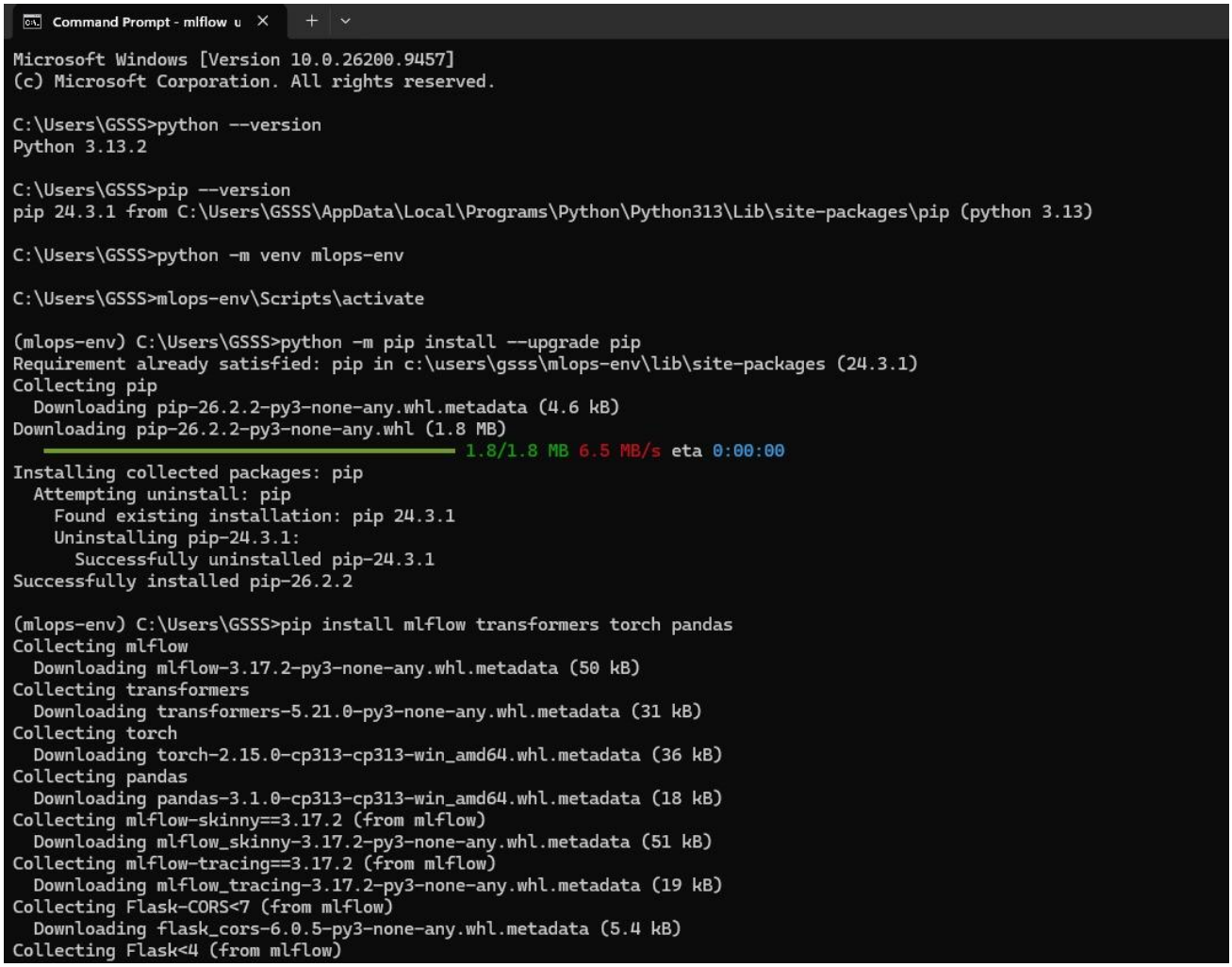

mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Description
No description
Metrics (3)
Search metrics
Metric
Value
accuracy
1
avg_confidence
0.9996546626091003
total_latency_sec
0.18375468254089355
Parameters (3)
Search parameters
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
8
num_samples
10
About this run
Created at
10/29/2026, 05:57:37 AM
Created by
GSSS
Experiment ID
1
Status
Finished
Run ID
cd352e7f61344399846b89e3c87aaa4a
Duration
345ms
Source
mlflow_lab.py
Logged models
—
Registered prompts
—
Datasets
None
Tags
Add tags
Registered models
None
Activate Windows
Go to Settings to activate Windows.
27°C
Sunny
Search
ENG
IN
28 October 2026
Wed 21:05 (Local time)

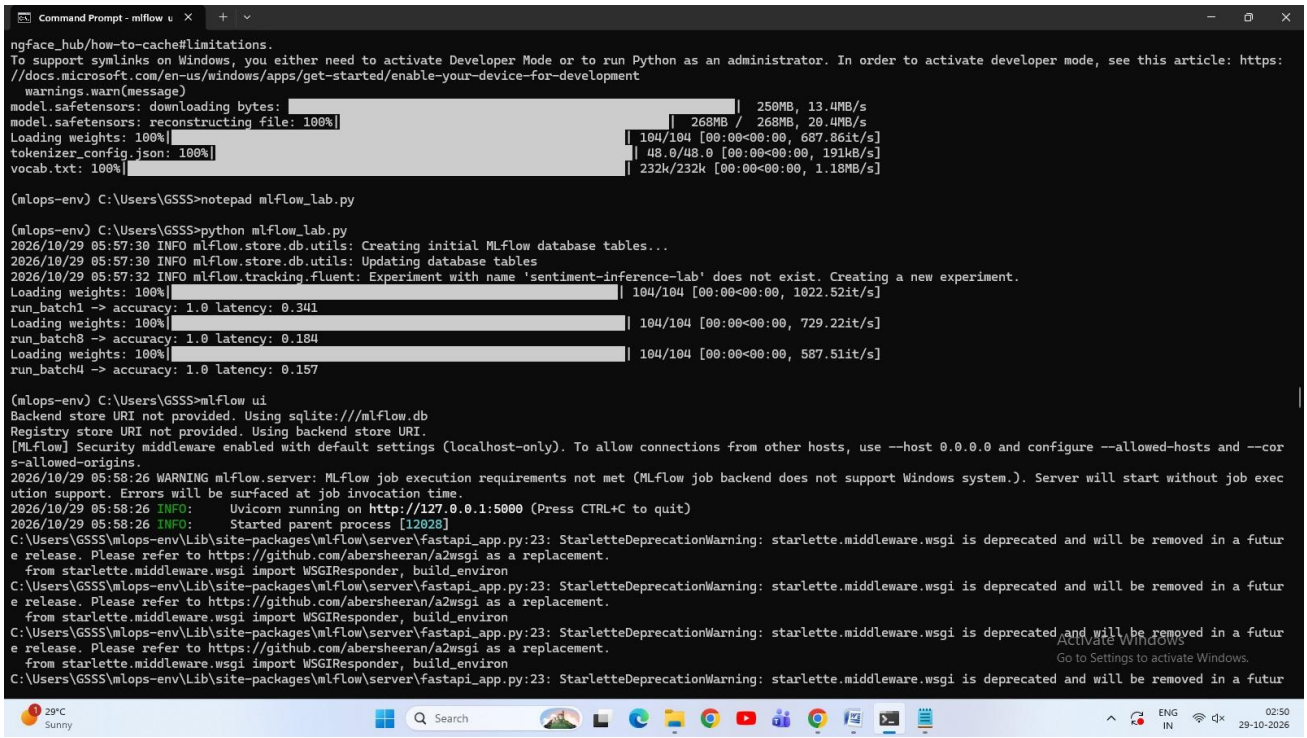

mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
1.00
1.00
0.18
0
0.2
0.4
0.6
0.8
1
0
0.05
0.1
0.15
run_batch8
Add section
Activate Windows
Go to Settings to activate Windows.
27°C
Sunny
Search
ENG
IN
21:06
28-10-2026


mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch1
Overview
Model metrics
System metrics
Traces
Artifacts
Description
No description
Metrics (3)
Search metrics
Metric
Value
accuracy
1
avg_confidence
0.9996546626091003
total_latency_sec
0.34099721908569336
Parameters (3)
Search parameters
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
1
num_samples
10
About this run
Created at
10/29/2026, 05:57:35 AM
Created by
GSSS
Experiment ID
1
Status
Finished
Run ID
ab21a34f61794f7b835b481d69d0b160
Duration
0.7s
Source
mlflow_lab.py
Logged models
—
Registered prompts
—
Datasets
None
Tags
Add tags
Registered models
None
Activate Windows
Go to Settings to activate Windows.
Show desktop
27°C
Sunny
Search
ENG
IN
21:06
28-10-2026


mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch1
Overview
Model metrics
System metrics
Traces
Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
1.00
1.00
0.34
0
0.2
0.4
0.6
0.8
1
0.1
0.3
run_batch1
Add section
Activate Windows
Go to Settings to activate Windows.
27°C
Sunny
Search
ENG
IN
21:06
28-10-2026

Execution:


Command Prompt - mlflow u
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.
C:\Users\GSSS>python --version
Python 3.13.2
C:\Users\GSSS>pip --version
pip 24.3.1 from C:\Users\GSSS\AppData\Local\Programs\Python\Python313\Lib\site-packages\pip (python 3.13)
C:\Users\GSSS>python -m venv mlops-env
C:\Users\GSSS>mlops-env\Scripts\activate
(mlops-env) C:\Users\GSSS>python -m pip install --upgrade pip
Requirement already satisfied: pip in c:\users\gsss\mlops-env\lib\site-packages (24.3.1)
Collecting pip
Downloading pip-26.2.2-py3-none-any.whl.metadata (4.6 kB)
Downloading pip-26.2.2-py3-none-any.whl (1.8 MB)
1.8/1.8 MB 6.5 MB/s eta 0:00:00
Installing collected packages: pip
Attempting uninstall: pip
Found existing installation: pip 24.3.1
Uninstalling pip-24.3.1:
Successfully uninstalled pip-24.3.1
Successfully installed pip-26.2.2
(mlops-env) C:\Users\GSSS>pip install mlflow transformers torch pandas
Collecting mlflow
Downloading mlflow-3.17.2-py3-none-any.whl.metadata (50 kB)
Collecting transformers
Downloading transformers-5.21.0-py3-none-any.whl.metadata (31 kB)
Collecting torch
Downloading torch-2.15.0-cp313-cp313-win_amd64.whl.metadata (36 kB)
Collecting pandas
Downloading pandas-3.1.0-cp313-cp313-win_amd64.whl.metadata (18 kB)
Collecting mlflow-skinny==3.17.2 (from mlflow)
Downloading mlflow_skinny-3.17.2-py3-none-any.whl.metadata (51 kB)
Collecting mlflow-tracing==3.17.2 (from mlflow)
Downloading mlflow_tracing-3.17.2-py3-none-any.whl.metadata (19 kB)
Collecting Flask-CORS<7 (from mlflow)
Downloading flask_cors-6.0.5-py3-none-any.whl.metadata (5.4 kB)
Collecting Flask<4 (from mlflow)


Command Prompt - mlflow u
ngface_hub/how-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: https://docs.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-development
warnings.warn(message)
model.safetensors: downloading bytes: 250MB, 13.4MB/s
model.safetensors: reconstructing file: 100% 268MB / 268MB, 20.4MB/s
Loading weights: 100% 104/104 [00:00<00:00, 687.86it/s]
tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 191kB/s]
vocab.txt: 100% 232k/232k [00:00<00:00, 1.18MB/s]
(mlops-env) C:\Users\GSSS>notepad mlflow_lab.py
(mlops-env) C:\Users\GSSS>python mlflow_lab.py
2026/10/29 05:57:30 INFO mlflow.store.db.utils: Creating initial MLflow database tables...
2026/10/29 05:57:30 INFO mlflow.store.db.utils: Updating database tables
2026/10/29 05:57:32 INFO mlflow.tracking.fluent: Experiment with name 'sentiment-inference-lab' does not exist. Creating a new experiment.
Loading weights: 100% 104/104 [00:00<00:00, 1022.52it/s]
run_batch1 -> accuracy: 1.0 latency: 0.341
Loading weights: 100% 104/104 [00:00<00:00, 729.22it/s]
run_batch8 -> accuracy: 1.0 latency: 0.184
Loading weights: 100% 104/104 [00:00<00:00, 587.51it/s]
run_batch4 -> accuracy: 1.0 latency: 0.157
(mlops-env) C:\Users\GSSS>mlflow ui
Backend store URI not provided. Using sqlite:///mlflow.db
Registry store URI not provided. Using backend store URI.
[MLflow] Security middleware enabled with default settings (localhost-only). To allow connections from other hosts, use --host 0.0.0.0 and configure --allowed-hosts and --cors-allowed-origins.
2026/10/29 05:58:26 WARNING mlflow.server: MLflow job execution requirements not met (MLflow job backend does not support Windows system.). Server will start without job execution support. Errors will be surfaced at job invocation time.
2026/10/29 05:58:26 INFO: Uvicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
2026/10/29 05:58:26 INFO: Started parent process [12028]
C:\Users\GSSS\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\GSSS\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
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C:\Users\GSSS\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\GSSS\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a futur
Activate Windows
Go to Settings to activate Windows.
29°C Sunny
Search
ENG IN
02:50 29-10-2026

Mlflow_lab.py file :

